

Throughput: SPAR wins where aisles are scarce, and loses where they are not

Taller is better: tasks delivered per 1000 timesteps, so 149 means 149 tasks out of every 1000 steps of simulated time.

Each map shows the best SPAR configuration on that map, named under its bar, with a permutation test against plain PIBT. The two wins are ahead by 21-27% but land at $p = 0.06$ on five seeds; the two losses are significant. Both baselines are far behind everywhere.

